GATE 2024 Official Chemical Engineering (CH) Question Paper

General Instructions:
1. This paper contains 65 questions.
2. Questions 1 to 35 carry 1 mark each. Questions 36 to 65 carry 2 marks each.

Q.1 For laminar flow of a Newtonian fluid in a circular pipe under steady state, the Fanning friction factor f is:
(A) $f = 16 / Re$
(B) $f = 64 / Re$
(C) $f = 0.079 / Re^{0.25}$
(D) $f = 24 / Re$

Q.2 Which of the following dimensionless numbers represent heat transfer ratio of conductive resistance to convective resistance?
(A) Reynolds Number
(B) Prandtl Number
(C) Nusselt Number
(D) Biot Number

Q.3 A continuous distillation column separates a binary mixture. The minimum reflux ratio is calculated as 1.5. Calculate the operating reflux ratio for 1.2 times R_{min} (round off to 2 decimal places).

Q.4 Which of the following statements is/are correct regarding ideal CSTR reactors? (Select all correct options)
(A) Fluid parameters are uniform throughout the reactor volume
(B) Exit stream composition is identical to the contents inside the reactor
(C) Space time equals volume divided by volumetric flow rate
(D) CSTR volume is always smaller than PFR volume for first-order reaction

Q.5 For an endothermic reaction A -> B operating at steady state, the equilibrium conversion increases with:
(A) Increasing Temperature
(B) Decreasing Temperature
(C) Increasing Pressure
(D) Addition of Inert Gas